Brandon Knutson

PhD Candidate in Applied Mathematics

Education

2022–Present **PhD in Applied Mathematics**, *Colorado School of Mines*, Golden, CO
Expected graduation: May 2026.

2014–2018 **Bachelor of Arts, with Distinction in Mathematics and Physics**, *University of Colorado Boulder*, Boulder, CO, *GPA: 3.99*
Dean's List every semester. Member of Phi Beta Kappa and Pi Mu Epsilon honor societies.

Research Interests

Optimization and machine learning applications, including implicit neural networks.

Research Projects

2023–Present **Randomized Jacobian-free Backpropagation**, *Advisors: Drs. Daniel McKenzie and Samy Wu Fung*, Colorado School of Mines
Developing an improvement to Jacobian-free backpropagation, a successful and efficient method for training implicit neural networks. The improved technique utilizes random forward iterations to train faster and with less memory, with inspiration drawn from the field of stochastic bilevel optimization.

2023–Present **Implicit Networks Applied to Maze Navigation**, *Advisors: Drs. Daniel McKenzie and Samy Wu Fung*, Colorado School of Mines
Investigating the extrapolation capabilities of implicit networks applied to two-dimensional mazes. Evaluating out-of-distribution performance of a benchmark model in extrapolation directions not previously explored. Identifying and quantifying periodic behavior within implicit networks using tools from Topological Data Analysis.

Publications

2026 **Brandon Knutson**, Michael Ivanitskiy, Jordan Pettyjohn, Cecilia Diniz Behn, Samy Wu Fung & Daniel McKenzie. *On Logical Extrapolation for Mazes with Recurrent and Implicit Networks.* Proceedings of the AAAI Conference on Artificial Intelligence (AAAI-26).

2025 Michael Ivanitskiy, Aaron Sandoval, Alexander F. Spies, Tilman Räuker, **Brandon Knutson**, Cecilia Diniz Behn & Samy Wu Fung. *maze-dataset: Maze Generation with Algorithmic Variety and Representational Flexibility.* Journal of Open Source Software, 10(114), 8633.

Research and Work Experience

2023 **Summer Intern**, *Oak Ridge National Laboratory*, Oak Ridge, TN
Constructed scalable model for estimating building heights from footprints and coarse-scale height data.

2019–2022 **Operations Research Analyst (GS-11)**, *US Navy*, Norfolk, VA
Attained secret security clearance. Developed cost models and analyzed federal acquisition proposals, securing significant savings.

2016–2017 **Physics Research Assistant**, *University of Colorado Boulder*, Boulder, CO
Developed a rapid thermal annealing system for use in an applied laser physics lab.

(720) 499-9097 • *brkn8102@colorado.edu*
https://bknutson0.github.io/ • *brandon-knutson-8102*
https://github.com/bknutson0

Teaching Experience

2024 **Instructor (Differential Equations)**, *Colorado School of Mines*, Golden, CO
Delivered concise, engaging lectures to facilitate effective student learning. Assisted with course design and grading.

2022–2023 **Teaching Assistant (Differential Equations)**, *Colorado School of Mines*, Golden, CO
Led recitations, held office hours, and graded assignments and tests.

2017–2018 **Math and Physics Tutor**, *University of Colorado Boulder*, Boulder, CO
Tutored algebra, calculus, analysis, and physics at the University of Colorado Boulder's Mathematics Academic Resource Center.

Presentations and Posters

March 2024 ***Extrapolating with a Recurrent Neural Network on Mazes***, *Poster presented at the Institute for Mathematical and Statistical Innovation*, Chicago, IL
Presented research on the extrapolation capabilities of a recurrent neural network trained to solve mazes with out-of-distribution shifts.

January 2025 ***Logical Extrapolation for Mazes with Recurrent and Implicit Networks***, *Poster presented at the U.S. Dynamics Days 2025*, Denver, CO
Presented research on out-of-distribution behavior and latent dynamics of recurrent and implicit neural networks.

April 2025 ***Generalizing with neural networks on mazes***, *Presentation at the Graduate Research and Discovery Symposium*, Golden, CO
Presented work on generalizing with recurrent neural networks, including failures due to goal mis-generalization and improvements due to dataset diversification.

Skills

Programming Python, Mathematica, MATLAB, SQL, LaTeX
Tools Object-Oriented Design, Git, Linux environment

Awards

2018 Marlene Massaro Pratto and David Pratto Scholarship in Mathematics
2025 Travel Grant, U.S. Dynamics Days 2025

Professional Memberships

2022–Present Member, Society for Industrial and Applied Mathematics (SIAM)

Service

2023–Present Graduate Student Government Representative

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